1. Finding Mean Using Python

```
data = [10, 20, 30, 40, 50]
mean_val = sum(data) / len(data)
print("Mean:", mean_val)
```

```
Mean: 30.0
```

2. Finding Median Using Python

```
data = [40, 10, 30, 50, 20]
data.sort()
n = len(data)

if n % 2 == 0:
    median_val = (data[n // 2 - 1] + data[n // 2]) / 2
else:
    median_val = data[n // 2]

print("Median:", median_val)
```

```
Median: 30
```

3. Finding Mode Using Python

```
from collections import Counter

data = [1, 2, 2, 3, 4, 2, 5]
count = Counter(data)
mode_val = count.most_common(1)[0][0]

print("Mode:", mode_val)
```

```
Mode: 2
```

4. Finding Range Using Python

```
data = [15, 8, 22, 42, 10]
range_val = max(data) - min(data)

print("Range:", range_val)
```

```
Range: 34
```

5. Finding Variance Using Python (Population)

```
data = [2, 4, 4, 4, 5, 5, 7, 9]

mean_val = sum(data) / len(data)
variance_val = sum((x - mean_val) ** 2 for x in data) / len(data)

print("Variance:", variance_val)
```

```
Variance: 4.0
```

6. Finding Standard Deviation Using Pure Python

```
data = [2, 4, 4, 4, 5, 5, 7, 9]

mean_val = sum(data) / len(data)
variance_val = sum((x - mean_val) ** 2 for x in data) / len(data)
std_dev = variance_val ** 0.5

print("Standard Deviation:", std_dev)
```

```
Standard Deviation: 2.0
```

7. Using Built-in Statistics Module

```
import statistics

data = [12, 15, 15, 18, 22, 25]

print("Mean:", statistics.mean(data))
print("Median:", statistics.median(data))
print("Mode:", statistics.mode(data))
print("Variance:", statistics.variance(data))
print("Standard Deviation:", statistics.stdev(data))
```

```
Mean: 17.833333333333332
Median: 16.5
Mode: 15
Variance: 23.766666666666666
Standard Deviation: 4.875106836436168
```

8. Using NumPy Library

```
import numpy as np
data = np.array([10, 20, 20, 30, 40])
print("Mean:", np.mean(data))
print("Median:", np.median(data))
print("Variance (Sample):", np.var(data, ddof=1))
print("Standard Deviation:", np.std(data, ddof=1))
print("Range:", np.ptp(data))
```

```
Mean: 24.0
Median: 20.0
Variance (Sample): 130.0
Standard Deviation: 11.40175425099138
Range: 30
```

```
9. Using SciPy for Mode
```

```
from scipy import stats
data = [5, 2, 5, 3, 5, 4]
mode_result = stats.mode(data, keepdims=True)
print("Mode:", mode_result.mode[0])
print("Count:", mode_result.count[0])
```

```
Mode: 5
Count: 3
```

10. Using Pandas for Complete Summary

```
import pandas as pd
df = pd.DataFrame({'values': [10, 20, 30, 40, 50, 50]})
print("Mean:", df['values'].mean())
print("Median:", df['values'].median())
print("Mode:", df['values'].mode().tolist())
print("Variance:", df['values'].var())
print("Std Dev:", df['values'].std())
print("Range:", df['values'].max() - df['values'].min())
```

```
Mean: 33.333333333333336
Median: 35.0
Mode: [50]
Variance: 266.6666666666667
Std Dev: 16.32993161855452
Range: 40
```